

Program : Diploma in Polymer Technology	
Course Code : 6097	Course Title: Product Development Workshop
Semester : 6	Credits: 2.5
Course Category: Program Elective	
Periods per week: 4 (L:0, T:1, P:3)	Periods per semester: 60

Course Objectives:

- To develop skills on design and development of complex rubber products.
- To perform the manufacturing techniques in laboratory level and its possibility of extension to the society

Course Prerequisites:

Topic	Course code	Course name	Semester
Rubber compounding, moulding and curing		Technology of elastomers	3
Rubber mastication and compounding		Rubber technology workshop	3
Mixing and Processing machinery:- two roll mill, extruder, calendar, hydraulic press.		Polymer processing and machinery	4
Design and development of rubber compound		Compound design workshop	5

Course Outcomes:

On completion of the course, the student will be able:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Conduct the compression molding of rubber products	20	Applying
CO2	Perform the manufacture of extruded and calendared rubber products.	20	Applying
CO3	Conduct the production of assembled rubber product	18	Applying
	Lab Exam	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3				
CO2	3		3				
CO3	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Conduct the compression molding of complex rubber products.		
M1.01	Carry out the production of washer	4	Applying
M1.02	Carryout the production of chair bush	4	Applying
M1.03	Carry out the production of mounting pad	4	Applying
M1.04	Carry out the production of oil seal	4	Applying
M1.05	Carry out the production of Stable mat (hollow mat)	4	Applying
	Lab Exam1	1	
CO2	Perform the manufacture of extruded and calendered rubber products.		
M2.01	Perform the production of channels	7	Applying
M2.02	Perform the production of tread.	7	Applying
M2.03	Perform the production of cushion gum.	6	Applying
CO3	Conduct the production of assembled rubber product.		
M3.01	Perform the production of hose	6	Applying
M3.02	Perform the production of conveyor belt	6	Applying
M3.03	Perform the production of Hawaii chapels	6	Applying
	Lab Exam II	1	

Text / Reference:

T/R	Book Title/Author
T1	Rubber Technology And Manufacture- C.M.Blow (Publisher – Butterworth Hainemann Ltd, 1982 Edition)
R2	Rubber Technology - Maurice Morton (Publisher – Springer, 1999 Edition)
R3	Vanderbilt Rubber Hand book -R.T Vanderbilt (Publisher – R.T Vanderbilt, 1990 Edition)
R4	Training Manual - RTI, RRII, Kottayam
R5	Rubbery Materials and their compounds - J.A Brydson (Publisher –Elsevier Applied Science, London, 1988)

Online Resources:

Sl.No	Website Link
1	www.researchgate.net
2	www.scribd.com
3	www.iri.net.in
4	www.sciencedirect.com